Class Field Theory

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1 Introduction

Class field theory is the theory that describes the nature of abelian extensions of global and local fields. A toy example is $\mathbb{Q}(\zeta_N)/\mathbb{Q}$, where ζ_N is a primitive Nth root of unity. We have

$$\mathrm{Gal}(\mathbb{Q}(\zeta_N)/\mathbb{Q}) \cong (\mathbb{Z}/N\mathbb{Z})^\times.$$

For a prime p that does not divide N, the above isomorphism sends $\mathrm{Frob}_p \in \mathrm{Gal}(\mathbb{Q}(\zeta_N)/\mathbb{Q})$ to $p \bmod N$. This fact brings up a criterion for decomposition of primes, namely, for a given prime p that does not divide N,

$$p \equiv 1 \bmod N \iff p \text{ is totally decomposed in } \mathbb{Q}(\zeta_N).$$

The group $(\mathbb{Z}/N\mathbb{Z})^\times$ reflects the way the primes decompose in $\mathbb{Q}(\zeta_N)$, which stays outside $\mathbb{Q}$. The Galois group $\mathrm{Gal}(\mathbb{Q}(\zeta_N)/\mathbb{Q})$ is (mysteriously) "approximated" by an easy-to-understand group that is originally unrelated to Galois theory. Similarly to Galois theory, we have "correspondence".

In class field theory, we may consider K^{ab}, the maximal abelian extension of a given commutative field K. $\mathrm{Gal}\left(K^{\mathrm{ab}}/K\right)$ is "approximated" by an easy-to-understand group $\mathbb{Z}$ (resp. $K^\times$, resp. $C_K \coloneqq \mathbb{A}_K^\times/K^\times$) when K is a finite field (resp. local field, resp. global field).

2 Local Class Field Theory

Throughout this section, K is assumed to be a local field.

Theorem 2.1 (Main theorem)**.** (1) There exists a unique continuous homomorphism

$$\theta\colon K^\times \longrightarrow \mathrm{Gal}\left(K^{\mathrm{ab}}/K\right)$$

that satisfies the following conditions (i) and (ii):

(i) If L is a finite abelian extension of K, then θ induces an isomorphism of the quotient groups

$$K^\times / \operatorname{N}_{L/K} L^\times \xrightarrow{\cong} \operatorname{Gal}(L/K)$$

where $\operatorname{N}_{L/K}$ is the norm map.

(ii) (relationship to the theory of abelian extensions of finite fields) If K is a complete discrete valuation field with a finite residue field $\mathbb{F}_q$, then we have a commutative diagram:

$$\begin{array}{ccc} K^\times & \xrightarrow{\theta} & \operatorname{Gal}\left(K^{\mathrm{ab}}/K\right) \\ {\scriptstyle \mathrm{val}_K}\downarrow & & \downarrow \\ \mathbb{Z} & \xrightarrow{\theta} & \operatorname{Gal}\left(\mathbb{F}_q^{\mathrm{ab}}/\mathbb{F}_q\right) \end{array}$$

Here, val_K is the discrete valuation of K, and the map $\operatorname{Gal}\left(K^{\mathrm{ab}}/K\right) \to \operatorname{Gal}\left(\mathbb{F}_q^{\mathrm{ab}}/\mathbb{F}_q\right)$ is the composition

$$\operatorname{Gal}\left(K^{\mathrm{ab}}/K\right) \longrightarrow \operatorname{Gal}\left(K^{\mathrm{ur}}/K\right) \cong \operatorname{Gal}\left(\mathbb{F}_q^{\mathrm{ab}}/\mathbb{F}_q\right)$$

where K^{ur} is the largest unramified extension of K.

(2) $U \mapsto \theta^{-1}(U)$ is a bijection from the set of open subgroups of $\operatorname{Gal}\left(K^{\mathrm{ab}}/K\right)$ to the set of open subgroups of finite index of $K^\times$.

Let K^{sep} be a separable closure of K. Recall that $\operatorname{Gal}\left(K^{\mathrm{sep}}/K\right) = \varprojlim \operatorname{Gal}\left(L_i/K\right)$, where $(L_i : i \in I)$ is the family of all finite Galois extensions of K, is a profinite group. Define $\operatorname{Br}(K) := H^2\left(\operatorname{Gal}\left(K^{\mathrm{sep}}/K\right), \left(K^{\mathrm{sep}}\right)^\times\right) = \varinjlim H^2\left(\operatorname{Gal}(L_i/K), L_i^\times\right)$ to be the **Brauer group** of the field K. The definition does not depend on the choice of K^{sep}. (On the other hand, the equivalence classes of central simple algebras over K, under tensor product as multiplication, form a group. This group is isomorphic to the Brauer group.)

We have $H^2\left(\operatorname{Gal}\left(K^{\mathrm{ur}}/K\right), \left(K^{\mathrm{ur}}\right)^\times\right) = \operatorname{Br}(K)$. The valuation map $v : \left(K^{\mathrm{ur}}\right)^\times \to \mathbb{Z}$ defines an isomorphism $H^2\left(\operatorname{Gal}\left(K^{\mathrm{ur}}/K\right), \left(K^{\mathrm{ur}}\right)^\times\right) \to H^2\left(\widehat{\mathbb{Z}}, \mathbb{Z}\right)$. Consider the short exact sequence

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Q} \longrightarrow \mathbb{Q}/\mathbb{Z} \longrightarrow 0 \ .$$

This induces an isomorphism $H^1\left(\widehat{\mathbb{Z}}, \mathbb{Q}/\mathbb{Z}\right) = \operatorname{Hom}\left(\widehat{\mathbb{Z}}, \mathbb{Q}/\mathbb{Z}\right) \to H^2\left(\widehat{\mathbb{Z}}, \mathbb{Z}\right)$. Define $\gamma : \operatorname{Hom}\left(\widehat{\mathbb{Z}}, \mathbb{Q}/\mathbb{Z}\right) \to \mathbb{Q}/\mathbb{Z}$ by $\phi \mapsto \phi(1)$. The map $\mathrm{inv}_K : H^2\left(\operatorname{Gal}\left(K^{\mathrm{ur}}/K\right), \left(K^{\mathrm{ur}}\right)^\times\right) \to \mathbb{Q}/\mathbb{Z}$ is now defined by

$$H^2\left(\operatorname{Gal}\left(K^{\mathrm{ur}}/K\right), \left(K^{\mathrm{ur}}\right)^\times\right) \xrightarrow{v} H^2\left(\widehat{\mathbb{Z}}, \mathbb{Z}\right) \xrightarrow{\delta^{-1}} \operatorname{Hom}\left(\widehat{\mathbb{Z}}, \mathbb{Q}/\mathbb{Z}\right) \xrightarrow{\gamma} \mathbb{Q}/\mathbb{Z},$$

i.e.,

$$\mathrm{inv}_K = \gamma \circ \delta^{-1} \circ v.$$

Indeed it is an isomorphism. So we have defined an isomorphism $\mathrm{inv}_K \colon \mathrm{Br}(K) \to \mathbb{Q}/\mathbb{Z}$.

Let L/K be a finite extension of degree n. Then

$$\mathrm{inv}_L \circ \mathrm{Res}_{K/L} = n \cdot \mathrm{inv}_K \,.$$

In other words, the following diagram is commutative:

$$\begin{array}{ccc} \mathrm{Br}(K) & \xrightarrow{\mathrm{Res}_{K/L}} & \mathrm{Br}(L) \\ {\scriptstyle \mathrm{inv}_K}\downarrow & & \downarrow{\scriptstyle \mathrm{inv}_L} \\ \mathbb{Q}/\mathbb{Z} & \xrightarrow{\quad n \quad} & \mathbb{Q}/\mathbb{Z} \end{array}$$

Note that $H^2\left(\mathrm{Gal}(L/K), L^\times\right)$ is the kernel of $\mathrm{Res}_{K/L}$. Then it is cyclic of order n. More precisely, $H^2\left(\mathrm{Gal}(L/K), L^\times\right)$ is generated by the element $u_{L/K} \in \mathrm{Br}(K)$, the invariant of which is $1/n \in \mathbb{Q}/\mathbb{Z}$.

By Tate's theorem, the map $\alpha \mapsto \alpha \cdot u_{L/K}$ given by the cup-product is an isomorphism of $\widehat{H}^q\left(\mathrm{Gal}(L/K), \mathbb{Z}\right)$ onto $\widehat{H}^{q+2}\left(\mathrm{Gal}(L/K), L^\times\right)$. For the case $q = -2$, we have $\widehat{H}^q\left(\mathrm{Gal}(L/K), \mathbb{Z}\right) = H_1\left(\mathrm{Gal}(L/K), \mathbb{Z}\right) = \mathrm{Gal}(L/K)/\,\mathrm{Gal}'(L/K) = \mathrm{Gal}^{\mathrm{ab}}(L/K)$ and $\widehat{H}^0\left(\mathrm{Gal}(L/K), L^\times\right) = K^\times/\,\mathrm{N}_{L/K}\,L^\times$. That is, the cup-product by $u_{L/K}$ defines an isomorphism of $\mathrm{Gal}^{\mathrm{ab}}(L/K)$ onto $K^\times/\,\mathrm{N}_{L/K}\,L^\times$. Define $\theta = \theta_{L/K}$ to be the isomorphism of $K^\times/\,\mathrm{N}_{L/K}\,L^\times$ onto $\mathrm{Gal}^{\mathrm{ab}}(L/K)$. The map θ is called the **local reciprocity map** or the **norm residue symbol**. If $\alpha \in K^\times$ corresponds to $\overline{\alpha} \in K^\times/\,\mathrm{N}_{L/K}\,L^\times$, then we write $\theta_{L/K}\left(\overline{\alpha}\right) = (\alpha, L/K)$.

Let L/K be an unramified extension of degree n and let $\mathrm{Frob} \in \mathrm{Gal}(L/K)$ be the Frobenius element. Let $\alpha \in K^\times$ and let $\mathrm{val}_K(\alpha) \in \mathbb{Z}$ be its normalized valuation. Then $(\alpha, L/K) = \mathrm{Frob}^{\mathrm{val}_K(\alpha)}$. Consequently, let E/K be a finite abelian extension. Then the local reciprocity map $K^\times \to \mathrm{Gal}(E/K)$ maps $\mathcal{O}_K^\times$ onto the inertia subgroup T of $\mathrm{Gal}(E/K)$.

A subgroup M of $K^\times$ is called a **norm subgroup** if there exists a finite abelian extension L/K with $M = \mathrm{N}_{L/K}\,L^\times$. A subgroup M of $K^\times$ is a norm subgroup if and only if it satisfies the following two conditions:

(1) Its index $[K^\times : M]$ is finite.

(2) M is open in $K^\times$.

We have the following commutative diagram with exact rows:

$$\begin{array}{ccccccccc} 1 & \longrightarrow & \mathcal{O}_K^\times & \longrightarrow & K^\times & \xrightarrow{\mathrm{val}_K} & \mathbb{Z} & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \theta} & & \downarrow{\scriptstyle \theta} & & \downarrow{\scriptstyle \mathrm{id}} & & \\ 1 & \longrightarrow & I_K & \longrightarrow & \mathrm{Gal}\left(K^{\mathrm{ab}}/K\right) & \longrightarrow & \widehat{\mathbb{Z}} & \longrightarrow & 1, \end{array}$$

where $I_K = \mathrm{Gal}\left(K^{\mathrm{ab}}/K^{\mathrm{ur}}\right)$ is the inertia subgroup of $\mathrm{Gal}\left(K^{\mathrm{ab}}/K\right)$, and $\mathrm{Gal}\left(K^{\mathrm{ur}}/K\right)$ is identified with $\widehat{\mathbb{Z}}$.

We can now state two equivalent formulations of the existence theorem: The map $\theta\colon \mathcal{O}_K^\times \to I_K$ is an isomorphism. The topology induced on $\mathcal{O}_K^\times$ by the norm subgroup is the natural topology of $\mathcal{O}_K^\times$.

Finally we will introduce the formal multiplication in local fields, which can be used to prove the existence theorem. Let A be a commutative ring with 1 and let $F \in A[[X,Y]]$. We say that F is a **commutative formal group law** if

(1) $F(X, F(Y,Z)) = F(F(X,Y), Z)$;

(2) $F(0,Y) = Y$ and $F(X,0) = X$;

(3) there is a unique $G(X)$ such that $F(X, G(X)) = 0$;

(4) $F(X,Y) = F(Y,X)$;

(5) $F(X,Y) \equiv X + Y \pmod{\deg 2}$.

Take $A = \mathcal{O}_K$, let $F(X,Y)$ be a commutative formal group law defined over $\mathcal{O}_K$ and let $\mathfrak{m}_K \subset \mathcal{O}_K$ be the maximal ideal. Under $x * y = F(x,y)$, $\mathfrak{m}_K$ forms a group, and we denote this group by $F(\mathfrak{m}_K)$.

Let K be a local field, $q = \#k = \#\mathcal{O}_K/\mathfrak{m}_K$ and choose a uniformizing element $\pi \in \mathcal{O}_K$. Let $\mathfrak{F}_\pi$ be the set of formal power series f with:

(1) $f(X) \equiv \pi X \pmod{\deg 2}$;

(2) $f(X) \equiv X^q \pmod{\pi}$.

Let $f \in \mathfrak{F}_\pi$, by successive approximations, one can construct a unique formal group law F_f with coefficients in $\mathcal{O}_K$ for which f is an endomorphism. Let $f \in \mathfrak{F}_\pi$ and F_f the corresponding group law. Then for any $a \in \mathcal{O}_K$ there exists a unique $[a]_f \in \mathcal{O}_K[[X]]$ such that $[a]_f$ commutes with f and $[a]_f \equiv aX \pmod{\deg 2}$. Moreover $[a]_f$ is then an endomorphism of the group law

F_f. The map $a \mapsto [a]_f$ is an injective homomorphism of the ring $\mathcal{O}_K$ into the ring $\mathrm{End}(F_f)$. Let f and g be members of $\mathfrak{F}_\pi$. Then the corresponding group laws are isomorphic.

Let K be a local field and let π be a uniformizing element. Let $f \in \mathfrak{F}_\pi$, and let F_f be the corresponding group law. Let $M_f = F_f\left(\mathfrak{m}_{K^{\mathrm{sep}}}\right)$ be the group of points in the separable closure equipped with the group law deduced from F_f. Let $a \in \mathcal{O}_K$, $x \in M_f$ and put $ax = [a]_f x$, this defines a structure of an A-module on M_f. Let E_f be the torsion submodule of M_f; that is the set of elements of M_f killed by a power of π. Then the following statements hold:

(a) The torsion sub-module E_f is isomorphic (as an A-module) with K/A.

(b) Let $K_\pi = K(E_f)$ be the field generated by E_f over K. Then K_π is an abelian extension of K.

(c) Let u be a unit in $K^\times$. Then the element $\sigma_u = (u, K_\pi/K)$ of $G(K_\pi/K)$ acts on E_f via $\left[u^{-1}\right]_f$.

(d) The operation described in (c) defines an isomorphism $U_K \longrightarrow G(K_\pi/K)$.

(e) The norm residue symbol $(\pi, K_\pi/K)$ is 1.

(f) The fields K^{ur} and K_π are linearly disjoint and $K^{\mathrm{ab}} = K^{\mathrm{ur}} \cdot K_\pi$.

Using these facts, one can construct the reciprocity map and prove the existence theorem.

3 Global Class Field Theory

Throughout this section, K is assumed to be a global field.

Theorem 3.1 (Main theorem)**.** (1) Every abelian extension L/K satisfies the reciprocity law (i.e., there is an Artin map $\psi_{L/K}$).

(2) The Artin map $\psi_{L/K}$ is surjective with the kernel $K^\times \mathrm{N}_{L/K}\left(\mathbb{A}_L^\times\right)$ and hence induces an isomorphism of $C_K/\mathrm{N}_{L/K}(C_L)$ onto $\mathrm{Gal}(L/K)$.

(3) If $M \supset L \supset K$ are abelian extensions, then the diagram

$$\begin{array}{ccc} C_K/\mathrm{N}_{M/K}(C_M) & \xrightarrow{\psi_{M/K}} & \mathrm{Gal}(M/K) \\ \downarrow{\scriptstyle j} & & \downarrow{\scriptstyle \theta} \\ C_K/\mathrm{N}_{L/K}(C_L) & \xrightarrow{\psi_{L/K}} & \mathrm{Gal}(L/K) \end{array}$$

commutes (where θ is the usual map and j is the natural surjective map which exists because $\mathrm{N}_{M/K}\, C_M \subset \mathrm{N}_{L/K}\, C_L$).

(4) (Existence theorem.) For every open subgroup N of finite index in C_K there exists a unique abelian extension L/K (in a fixed algebraic closure of K) such that $\mathrm{N}_{L/K}\, C_L = N$.

Let L/K be a finite extension. Let $S \subset |K|$ be a finite set containing all the archimedean places and ramified places. We will denote by I^S the free abelian group on the elements of $|K| - S$. Assume now that L/K is abelian, $F_{L/K}$ maps every place $v \in |K| - S$ to $\mathrm{Frob}_v \in \mathrm{Gal}(L/K)$. By linearity we can extend this to a homomorphism (to be denoted by $F_{L/K}$ also) of I^S into $\mathrm{Gal}(L/K)$, putting

$$F_{L/K}\left(\sum_{v\notin S} n_v v\right) = \prod_{v\notin S} F_{L/K}(v)^{n_v},$$

where the n_v are integers and $n_v = 0$ except for a finite number of the v.

If $a \in K^\times$ (i.e., a is a non-zero element of K), then we write

$$(a)^S = \sum_{v\notin S} n_v v,$$

where $n_v = \mathrm{val}_v(a)$ for all $v \notin S$; thus $(a)^S$ is an element of I^S.

A crude form of the reciprocity law: if L/K is a finite abelian extension and S is the set of places of K consisting of the archimedean ones and those ramified in L, then there exists $\varepsilon > 0$ such that if $a \in K^\times$ and $|a-1|_v < \varepsilon$ for all $v \in S$, then $F\left((a)^S\right) = 1$. In other words, if $a \in K^\times$ is sufficiently near to 1 at all primes in a large enough set S, then $F\left((a)^S\right) = \prod_{v\notin S} F(v)^{\mathrm{val}_v(a)} = 1$.

Let G be a commutative topological group. A homomorphism $\phi\colon I^S \to G$ is said to be **admissible** if for each neighborhood N of the identity element 1 of G there exists $\varepsilon > 0$ such that $\phi\left((a)^S\right) \in N$ whenever $a \in K^\times$ and $|a-1|_v < \varepsilon$ for all $v \in S$.

One can prove that the reciprocity law holds for a finite abelian extension L of K if and only if there exists a continuous homomorphism ψ of $\mathbb{A}_K^\times \to G(L/K)$ such that

(i) ψ is continuous.

(ii) $\psi(K^\times) = 1$.

(iii) $\psi(x) = F_{L/K}((x)^S)$ for all $x \in J_K^S$, where S consists of the archimedean primes of K and those ramified in L.

Here the set of elements of the idèle group $\mathbb{A}_K^\times$ which have the value 1 at all the vth components, $v \in S$, is denoted by J_K^S. If $x \in \mathbb{A}_K^\times$ it has a non-unit component at only a finite number of

v-components; if the (additive) valuation of the vth component x_v of x is $n_v \in \mathbb{Z}$ we write

$$(x)^S = \sum_{v \notin S} n_v v \in I^S.$$

Such a map $\psi = \psi_{L/K}$ is called the **Artin map** associated with the extension L/K. Since it acts trivially on $K^\times$, it may be viewed as a map of the idèle class group $C_K = \mathbb{A}_K^\times / K^\times$ into $\mathrm{Gal}(L/K)$.

Finally we will introduce the relation between global and local Artin maps. Let L/K be an abelian extension. Let v be a place of K and w be a place of L lying over v. We have

$$K_v^\times \underset{j_v}{\overset{i_v}{\rightleftarrows}} \mathbb{A}_K^\times \xrightarrow{\psi_{L/K}} \mathrm{Gal}(L/K)$$

where i_v is the mapping of an $x \in K_v^\times$ onto the element of $\mathbb{A}_K^\times$ whose v component is x and whose other components are 1, and j_v is the projection onto the vth component. Call $\psi_v = \psi_{L/K} \circ i_v$; so $\psi_v \colon K_v^\times \to \mathrm{Gal}(L/K)$. In fact, $\psi_v \left(\mathrm{N}_{L_w/K_v} (L_w)^\times \right) = 1$. We shall call $\psi_v \colon K_v^\times \to \mathrm{Gal}(L_w/K_v)$ the **local Artin map**, or **norm residue homomorphism**. We have

$$\psi_{L/K}(x) = \prod_v \psi_v(x_v).$$

Indeed, the local Artin map coincides with the local reciprocity map defined in the local class field theory.

We can now give an apparently stronger statement of the reciprocity law. Let L/K be abelian, and let S consist of the archimedean primes of K and those ramified in L. If an element $a \in K^\times$ is a norm from L_w for all $v \in S$, then $F_{L/K} \left((a)^S \right) = 1$.

Now we give a cohomological proof of global class field theory.

To begin let's review some facts of cohomology of local fields. If L/K is finite unramified Galois extension of local fields, then $\widehat{H}^r \left(\mathrm{Gal}(L/K), \mathcal{O}_L^\times \right) = 0$ for all r. Let L/K be a cyclic extension of local fields of degree n, then we have the Herbrand quotient $h \left(\mathcal{O}_L^\times \right) = 1$ and $h \left(L^\times \right) = n$. Let A be an abelian group and let n be an integer ≥ 1, define

$$h_n(A) = \frac{\#(A/nA)}{\#_n A}$$

where ${}_nA$ is the set of $\alpha \in A$ such that $n\alpha = 0$. Let K be a local field and let $n \geq 1$ be an integer prime to the characteristic of K, then $h_n \left(K^\times \right) = n/|n|_K$.

We return to our proof of global class field theory. Let L/K be a finite Galois extension. Let $v \in |K|$ and $w \in |L|$ lying over v. Note that

$$\mathrm{Hom}_{G_{w_0}} \left(\mathbb{Z}[G], L_{w_0}^\times \right) = \prod_{G/G_{w_0}} L_{w_0}^\times = \prod_{w|v} L_w^\times,$$

We have mutually inverse isomorphisms

$$H^r\left(\mathrm{Gal}(L/K), \prod_{w|v} L_w^\times\right) \underset{j_{w_0}\,.\mathrm{res}}{\overset{\mathrm{cores.}\, i_{w_0}}{\longleftrightarrow}} H^r\left(G_{w_0}, L_w^\times\right)$$

and

$$H^r\left(\mathrm{Gal}(L/K), \prod_{w|v} \mathcal{O}_w^\times\right) \underset{j_{w_0}\,.\mathrm{res}}{\overset{\mathrm{cores.}\, i_{w_0}}{\longleftrightarrow}} H^r\left(G_{w_0}, \mathcal{O}_{w_0}^\times\right)$$

by Shapiro's lemma.

If $S \subset |K|$ contains all the ramified primes, $\prod_{v\notin S}\prod_{w|v} U_w$ has trivial cohomology. Hence

$$\widehat{H}^r\left(\mathrm{Gal}(L/K), J_{L,S}\right) \simeq \prod_{v\in S} \widehat{H}^r\left(G^v, (L^v)^\times\right)$$

where $L^v = L_w$ for arbitrary $w \mid v$ by the above assertion, and

$$J_{L,S} = \prod_{v\in S}\prod_{w|v} L_w^\times \times \prod_{v\notin S}\prod_{w|v} \mathcal{O}_w^\times.$$

Let $S \to |K|$ and take direct limit, we conclude

$$\widehat{H}^r\left(\mathrm{Gal}(L/K), \mathbb{A}_L^\times\right) \simeq \bigoplus_{v\in|K|} \widehat{H}^r\left(G^v, (L^v)^\times\right).$$

Consequently by Hilbert 90 and facts of local case,

$$H^1\left(\mathrm{Gal}(L/K), \mathbb{A}_L^\times\right) = 0,$$

$$H^2\left(\mathrm{Gal}(L/K), \mathbb{A}_L^\times\right) = \bigoplus_v \left(\frac{1}{n_v}\mathbb{Z}/\mathbb{Z}\right), \quad \text{where } n_v = [L^v : K_v].$$

Theorem 3.2. Let L/K be a cyclic extension of degree n. Then the Herbrand quotient

$$h(\mathrm{Gal}(L/K), C_L) \coloneqq \frac{\#H^2(\mathrm{Gal}(L/K), C_L)}{\#H^1(\mathrm{Gal}(L/K), C_L)} = n.$$

Sketch of proof. Let $S \subset |K|$ include the archimedean places, ramified places and all primes of K lying below some primes whose classes generate the ideal class group of L. Then $\mathbb{A}_L^\times = L^\times \cdot J_{L,S}$.

Denote by T the places of L which are above the primes in S. Hence

$$C_L = \mathbb{A}_L^\times / L^\times \cong J_{L,S} / \left(L^\times \cap J_{L,S}\right) = J_{L,S}/L_T$$

where $L_T = L^\times \cap J_{L,S}$ is the set of T-units of L. Then

$$h(C_L) = h(J_{L,S})/h(L_T).$$

But

$$h(J_{L,S}) = h\left(\prod_{v\in S}\prod_{w|v} L_w^\times\right) = \prod_{v\in S} h\left(\prod_{w|v} L_w^\times\right) = \prod_{v\in S} n_v$$

where the n_v are local degrees. So it only suffices to show

$$nh(L_T) = \prod_{v\in S} n_v.$$

One can construct a real vector space, on which G operates, with two lattices such that one has Herbrand quotient $nh(L_T)$ and the other has quotient $\prod_{v\in S} n_v$. □

Consequently, from $H^2\left(\mathrm{Gal}(L/K), C_L\right) = \mathbb{A}_K^\times/K^\times \mathrm{N}_{L/K}\,\mathbb{A}_L^\times$, we conclude

$$\left[\mathbb{A}_K^\times/K^\times \mathrm{N}_{L/K}\,\mathbb{A}_L^\times\right] \geq n$$

It is called the **first inequality**.

If L/K is a finite abelian extension and D is a subgroup of $\mathbb{A}_K^\times$ such that $D \subset \mathrm{N}_{L/K}\,\mathbb{A}_L^\times$ and $K^\times D$ is dense in $\mathbb{A}_K^\times$, then $L = K$.

If S is a finite subset of $|K|$ and L/K is a finite abelian extension, then $\mathrm{Gal}(L/K)$ is generated by the elements $F_{L/K}(v)$ for $v \notin S$, i.e., the map $F_{L/K}\colon I^S \to \mathrm{Gal}(L/K)$ is surjective. Hence, if L is a non-trivial abelian extension of K, there are infinitely many primes v of K that do not split completely (i.e., for which $F_{L/K}(v) \neq 1$).

Theorem 3.3. Let L/K be a Galois extension of degree n, with Galois group $\mathrm{Gal}(L/K)$. Then

1. $\left[\widehat{H}^0(\mathrm{Gal}(L/K), C_L)\right]$ and $\left[\widehat{H}^2(\mathrm{Gal}(L/K), C_L)\right]$ divide n.
2. $\widehat{H}^1(\mathrm{Gal}(L/K), C_L) = 0$.

Sketch of proof. We can reduce to case when n is prime and K contains the nth root of unity. In fact we shall prove:

Let K contain the nth roots of unity and L/K be an abelian extension of prime exponent n, with say $\mathrm{Gal}(L/K) \simeq (\mathbb{Z}/n\mathbb{Z})^r$. *Then*

$$[C_K : \mathrm{N}_{L/K}\, C_L] \textit{ divides } [L : K] = n^r.$$

By Kummer theory we know that $L = K\left(\sqrt[n]{a_1}, \cdots, \sqrt[n]{a_r}\right)$ for some $a_1, \cdots, a_r \in K$. Take S to be a finite set of (bad) primes, such that

(i) S contains all archimedean places,

(ii) S contains all divisors of n,

(iii) $J_K = K^\times \cdot J_{K,S}$ (by making S contain representatives of a system of generators for the ideal class group),

(iv) S contains all factors of the numerator and denominator of any a_i.

Write $M = K\left(\sqrt[n]{K_S}\right)$ for the field obtained from K by adjoining nth roots of all S-units. By the unit theorem K_S has a finite basis, so this extension is finite. Now $M \supset L \supset K$ and

$$K_S = {M^\times}^n \cap K_S \supset {L^\times}^n \cap K_S \supset {K^\times}^n \cap K_S = K_S^n.$$

By Kummer theory with $[M : L] = n^t$, $[L : K] = n^r$ (given) and $[M : K] = n^s$ we have

$$\left[K_S : {L^\times}^n \cap K_S\right] = n^t, \quad \left[{L^\times}^n \cap K_S : K_S^n\right] = n^r, \quad [K_S : K_S^n] = n^s$$

respectively. We have $s = \#S$.

If w is a prime of L above a $v \notin S$, then $F_{M/L}(w)$ is well-defined. Choose $w_1, \cdots, w_t$ so that $F_{M/L}(w_i)$ $(i = 1, \cdots, t)$ are a basis for $G = \mathrm{Gal}(M/L)$, and let $v_1, \cdots, v_t$ be primes of K below them. Then v_i is totally decomposed in L. We have $G_{v_i} = G_{w_i}$ and $L_{w_i} = K_{v_i}$ for all $i = 1, \cdots, t$.

Let

$$E = \prod_{v \in S} {K_v^\times}^n \times \prod_{v \in T} K_v^\times \times \prod_{v \notin S \cup T} \mathcal{O}_v^\times.$$

So $E \subset J_{K,S\cup T} \cap \mathrm{N}_{L/K}\, \mathbb{A}_L^\times$.

Now

$$\left[C_K / \mathrm{N}_{L/K}\, C_L\right] = \left[\mathbb{A}_K^\times / K^\times \mathrm{N}_{L/K}\, \mathbb{A}_L^\times\right]$$

divides $\left[\mathbb{A}_K^\times : K^\times E\right]$ because $E \subset \mathrm{N}_{L/K}\, \mathbb{A}_L^\times$. Combine with $\mathbb{A}_K^\times = K^\times J_{K,S} = K^\times J_{K,S\cup T}$, to prove $[C_K : \mathrm{N}_{L/K}\, C_L]$ divides $n^r = \left[{L^\times}^n \cap K_S : K_S^n\right]$, it suffices to show

$$[J_{K,S\cup T} : E] \,/\, [K_{S\cup T} : K \cap E] = n^r$$

(where $K_{S\cup T} = K^\times \cap J_{K,S\cup T}$).

First we calculate

$$[J_{K,S\cup T} : E] = \prod_{v \in S} \left[K_v^\times : \left(K_v^\times\right)^n\right] = n^{2s} \prod_{v \in S} |n|_v^{-1} = n^{2s}.$$

Thus it will be enough to show that

$$\left[K_{S\cup T} : K^\times \cap E\right] = n^{s+t}.$$

So it will be enough to show that $K^\times \cap E = K^n_{S\cup T}$. Clearly $K^\times \cap E \supset K^n_{S\cup T}$, so it remains to prove $K^\times \cap E \subset K^n_{S\cup T}$. This will result from the following lemma:

Let K contain the nth root of unity. Let S be a subset of $|K|$ satisfying (i), (ii), (iii) above, and let T be a set of primes disjoint from S, and independent for K_S in the sense that the map $K_S \to \prod_{v\in T} \mathcal{O}_v^\times / (\mathcal{O}_v^\times)^n$ is surjective. Suppose that $b \in K^\times$ is an nth power in S, arbitrary in T, and a unit outside $S \cup T$. Then $b \in {K^\times}^n$. □

As a consequence, if L/K is abelian with Galois group G, and there is an Artin map $\psi\colon \widehat{H}^0(\mathrm{Gal}(L/K), C_L) = C_K/\,\mathrm{N}_{L/K}\, C_L \to \mathrm{Gal}(L/K)$, then ψ must be an isomorphism.

Let n be a prime and let K be a field, not of characteristic n, containing the nth roots of unity. Let S be a finite set of primes of K satisfying the conditions (i), (ii), (iii) above, and let $M = K\left(\sqrt[n]{K_S}\right)$. Then, if the reciprocity law holds for M/K, we have

$$K^\times \,\mathrm{N}_{M/K}\, \mathbb{A}_M^\times = K^\times E, \text{ where } E = \prod_{v\in S} \left(K_v^\times\right)^n \times \prod_{v\notin S} \mathcal{O}_v^\times.$$

Let L/K be a finite (not necessarily abelian) Galois extension. From the short exact sequence

$$0 \longrightarrow L^\times \longrightarrow \mathbb{A}_L^\times \longrightarrow C_L \longrightarrow 0$$

we get exact sequence

$$0 \longrightarrow H^2\left(\mathrm{Gal}(L/K), L^\times\right) \longrightarrow \bigoplus_{v\in|K|} H^2\left(G^v, (L^v)^\times\right)$$

This gives the **Brauer–Hasse–Noether Theorem**: a central simple algebra over K splits over K if and only if it splits locally everywhere.

More precisely, if we regard the Brauer group as the equivalence classes of central simple algebras, under tensor product as multiplication,

Theorem 3.4 (Brauer–Hasse–Noether)**.** Let K be a global field.

(1) If $\alpha \in \mathrm{Br}(K)$, then for almost all places v the image α_{K_v} by the canonical map $\mathrm{Br}(K) \to \mathrm{Br}(K_v)\colon [A] \mapsto [A \otimes_K K_v]$ is 0. In other words, the image of the homomorphism $\mathrm{Br}(K) \to \prod_v \mathrm{Br}(K_v)\colon \alpha \mapsto (\alpha_{K_v})$ lies in the direct sum

$$\bigoplus_v \mathrm{Br}(K_v) = \left\{ (\alpha_v)_v \in \prod_v \mathrm{Br}(K_v) \mid \alpha_v = 0 \text{ for all but finitely many } v \right\}.$$

(2) The sequence

$$0 \longrightarrow \mathrm{Br}(K) \longrightarrow \bigoplus_v \mathrm{Br}(K_v) \longrightarrow \mathbb{Q}/\mathbb{Z} \longrightarrow 0$$

is exact, where the third arrow is the map

$$(\alpha_v)_v \longmapsto \sum_v \mathrm{inv}_{K_v}(\alpha_v).$$

We return to the cohomological proof of global class field theory, and prove the reciprocity law. To prove the reciprocity law, it will be convenient to state two related theorems, and to prove them both at once, gradually extending the cases for which they are true.

Theorem 3.5. Every finite abelian extension L/K satisfies the reciprocity law, and the Artin map $\theta\colon \mathbb{A}_K^\times \to \mathrm{Gal}(L/K)$ is given by product of local reciprocity maps $\theta = \prod_v \theta_v$.

Theorem 3.6. If $\alpha \in \mathrm{Br}(K)$, then $\sum_{v\in|K|} \mathrm{inv}_v(\alpha) = 0$.

Note that Theorem 3.5 is equivalent to

$$\theta(a) = \prod_{v\in|K|} \theta_v(a) = 1 \text{ for all } a \in K^\times.$$

If $\alpha \in \mathrm{Br}(K)$, then α is split by a finite extension of K means $\alpha \in H^2\left(\mathrm{Gal}(L/K), L^\times\right)$ for some finite extension L/K.

The proof of reciprocity law is in four main steps.

Step 1. Prove Theorem 3.5 for an arbitrary finite cyclotomic extension L/K.

Step 2. Deduce Theorem 3.6 for α split by a cyclic cyclotomic extension L/K.

Step 3. Deduce Theorem 3.6 for arbitrary $\alpha \in \mathrm{Br}(K)$.

Step 4. Deduce Theorem 3.5 for all abelian extension.

Steps 2 and 4 follow from the following identity for each v:

$$\mathrm{inv}_v\left(\overline{x}\cdot\delta\chi\right) = \chi_v\left(\theta_v(x_v)\right)$$

and so

$$\sum_v \mathrm{inv}_v\left(\overline{x}\cdot\delta\chi\right) = \chi(\theta(x)).$$

Here, $\chi \in \mathrm{Hom}(G, \mathbb{Q}/\mathbb{Z}) = H^1\left(\mathrm{Gal}(L/K), \mathbb{Q}/\mathbb{Z}\right)$, and for $v \in |K|$, χ_v is the restriction of χ to G^v, δ is the connecting homomorphism $H^1\left(\mathrm{Gal}(L/K), \mathbb{Q}/\mathbb{Z}\right) \to H^2\left(\mathrm{Gal}(L/K), \mathbb{Z}\right)$.

For Step 1 we reduce to case $K = \mathbb{Q}$. Let $\mathbb{Q}^{\mathrm{mc}}$ be the maximal cyclotomic extension of $\mathbb{Q}$. It only suffices to show that the local maps $\psi_p\colon \mathbb{Q}_p^\times \to \mathrm{Gal}\left(\mathbb{Q}_p^{\mathrm{mc}}/\mathbb{Q}_p\right)$ (by direct computation) are the same as θ_p (local Artin map) for all primes p.

For Step 3, It is enough to show that every element of $\mathrm{Br}(K)$ has a cyclic, cyclotomic splitting field. In other words, for every $\alpha \in \mathrm{Br}(K)$, there is a cyclic, cyclotomic extension L/K such that for every $v \in |K|$, the local degree $[L^v : K_v]$ is a multiple of the denominator of $\mathrm{inv}_v(\alpha)$. So we need only prove the lemma:

Given a number field K of finite degree over $\mathbb{Q}$, a finite set of primes S of K, and a positive integer m, there exists a cyclic, cyclotomic extension L/K whose local degrees are divisible by m at the non-archimedean primes v of S and divisible by 2 at real archimedean primes v of S (in other words, L is complex).

We now prove the third inequality. Let $E/L/K$ be finite Galois extensions of K; then we have an exact commutative diagram

$$\begin{array}{ccccccc}
 & & 0 & & 0 & & 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
0 & \longrightarrow & H^2\left(\mathrm{Gal}(L/K), L^\times\right) & \longrightarrow & H^2\left(\mathrm{Gal}(L/K), \mathbb{A}_L^\times\right) & \longrightarrow & H^2\left(\mathrm{Gal}(L/K), C_L\right) \\
 & & \downarrow & & \downarrow & & \downarrow \\
0 & \longrightarrow & H^2\left(\mathrm{Gal}(E/K), E^\times\right) & \longrightarrow & H^2\left(\mathrm{Gal}(E/K), \mathbb{A}_E^\times\right) & \longrightarrow & H^2\left(\mathrm{Gal}(E/K), C_E\right) \\
 & & \downarrow & & \downarrow & & \downarrow \\
0 & \longrightarrow & H^2\left(\mathrm{Gal}(E/L), E^\times\right) & \longrightarrow & H^2\left(\mathrm{Gal}(E/L), \mathbb{A}_E^\times\right) & \longrightarrow & H^2\left(\mathrm{Gal}(E/L), C_E\right)
\end{array}$$

In this diagram, the vertical lines are inflation-restriction sequences; they are exact since $H^1\left(\mathrm{Gal}(E/L), E^\times\right) = 0$ (Hilbert 90), $H^1\left(\mathrm{Gal}(E/L), \mathbb{A}_E^\times\right) = 0$ and $H^1\left(\mathrm{Gal}(E/L), C_E\right) = 0$ (Theorem 3.3). The horizontal sequences are exact since they come from the short exact sequence

$$0 \longrightarrow L^\times \longrightarrow \mathbb{A}_L^\times \longrightarrow C_L \longrightarrow 0$$

and the fact $H^1\left(\mathrm{Gal}(L/K), C_L\right) = 0$, etc. We pass to the limit and let $E \to \overline{K}$, the algebraic closure of K, to obtain the new commutative diagram

$$\begin{array}{ccccccc}
 & & 0 & & 0 & & 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
0 & \longrightarrow & H^2\left(\mathrm{Gal}(L/K), L^\times\right) & \xrightarrow{\gamma_1} & H^2\left(\mathrm{Gal}(L/K), \mathbb{A}_L^\times\right) & \xrightarrow{\varepsilon_1} & H^2\left(\mathrm{Gal}(L/K), C_L\right) \\
 & & \downarrow & & \downarrow & & \downarrow \\
0 & \longrightarrow & H^2\left(\mathrm{Gal}\left(\overline{K}/K\right), \overline{K}^\times\right) & \xrightarrow{\gamma_2} & H^2\left(\mathrm{Gal}\left(\overline{K}/K\right), \mathbb{A}_{\overline{K}}^\times\right) & \xrightarrow{\varepsilon_2} & H^2\left(\mathrm{Gal}\left(\overline{K}/K\right), C_{\overline{K}}\right) \\
 & & \downarrow & & \downarrow & & \downarrow \\
0 & \longrightarrow & H^2\left(\mathrm{Gal}\left(\overline{K}/L\right), \overline{K}^\times\right) & \xrightarrow{\gamma_3} & H^2\left(\mathrm{Gal}\left(\overline{K}/L\right), \mathbb{A}_{\overline{K}}^\times\right) & \xrightarrow{\varepsilon_3} & H^2\left(\mathrm{Gal}\left(\overline{K}/L\right), C_{\overline{K}}\right)
\end{array}$$

We are going to enlarge the above commutative diagram. For the Galois extension L/K, Theorem 3.6 ($\sum_{v\in|K|}\operatorname{inv}_v(\alpha)=0$ for $\alpha\in\operatorname{Br}(L/K)\subset\operatorname{Br}(K)$) tells us that the sequence

$$0\longrightarrow H^2\left(\operatorname{Gal}(L/K),L^\times\right)\xrightarrow{\gamma_1}H^2\left(\operatorname{Gal}(L/K),\mathbb{A}_L^\times\right)\xrightarrow{\operatorname{inv}_1}\mathbb{Q}/\mathbb{Z}$$

is a complex. Since $\operatorname{inv}_v(\operatorname{infl}\alpha)=\operatorname{inv}_v(\alpha)$ for all $\alpha\in H^2\left(\operatorname{Gal}(L/K),\mathbb{A}_L^\times\right)$, we have the following commutative diagram

$$\begin{array}{ccc} H^2\left(\operatorname{Gal}(L/K),\mathbb{A}_L^\times\right) & \xrightarrow{\operatorname{inv}_1} & \mathbb{Q}/\mathbb{Z} \\ {\scriptstyle\operatorname{infl}}\downarrow & & \| \\ H^2\left(\operatorname{Gal}\left(\overline{K}/K\right),\mathbb{A}_{\overline{K}}^\times\right) & \xrightarrow{\operatorname{inv}_2} & \mathbb{Q}/\mathbb{Z} \end{array}$$

Furthermore the sequence

$$0\longrightarrow H^2\left(\operatorname{Gal}\left(\overline{K}/K\right),\overline{K}^\times\right)\xrightarrow{\gamma_2}H^2\left(\operatorname{Gal}\left(\overline{K}/K\right),\mathbb{A}_{\overline{K}}^\times\right)\xrightarrow{\operatorname{inv}_2}\mathbb{Q}/\mathbb{Z}$$

is a complex. In a similar manner we have a complex

$$0\longrightarrow H^2\left(\operatorname{Gal}\left(\overline{K}/L\right),\overline{K}^\times\right)\xrightarrow{\gamma_3}H^2\left(\operatorname{Gal}\left(\overline{K}/L\right),\mathbb{A}_{\overline{K}}^\times\right)\xrightarrow{\operatorname{inv}_3}\mathbb{Q}/\mathbb{Z}\ .$$

Since $\operatorname{inv}_w(\operatorname{res}\alpha)=n_{w/v}\operatorname{inv}_v(\alpha)$, where $\alpha\in H^2\left(\operatorname{Gal}\left(\overline{K}/K\right),\mathbb{A}_{\overline{K}}^\times\right)$ and w is a prime of L over v of K, and $n_{w|v}=[L_w:K_v]$. One computes that

$$\sum_{w\in|L|}\operatorname{inv}_v(\operatorname{res}\alpha)=\sum_{v\in|K|}\sum_{w|v}n_{w|v}\operatorname{inv}_v(\alpha)=n\sum_{v\in|K|}\operatorname{inv}_v(\alpha).$$

That is, we have the commutative diagram

$$\begin{array}{ccc} H^2\left(\operatorname{Gal}\left(\overline{K}/K\right),\mathbb{A}_{\overline{K}}^\times\right) & \xrightarrow{\operatorname{inv}_2} & \mathbb{Q}/\mathbb{Z} \\ {\scriptstyle\operatorname{res}}\downarrow & & \downarrow{\scriptstyle n} \\ H^2\left(\operatorname{Gal}\left(\overline{K}/L\right),\mathbb{A}_{\overline{K}}^\times\right) & \xrightarrow{\operatorname{inv}_3} & \mathbb{Q}/\mathbb{Z} \end{array}$$

as the sum of the local degrees $\sum_{w|v}n_{w|v}=n=[L:K]$.

Now let $H^2\left(\operatorname{Gal}\left(\overline{K}/K\right),C_{\overline{K}}\right)_{\text{reg}}:=\operatorname{Im}(\varepsilon_2)$ and $H^2\left(\operatorname{Gal}\left(\overline{K}/L\right),C_{\overline{K}}\right)_{\text{reg}}:=\operatorname{Im}(\varepsilon_3)$. It follows that we have a map β_2 (resp. β_3) induced by inv_2 (resp. inv_3) of $H^2\left(\operatorname{Gal}\left(\overline{K}/K\right),C_{\overline{K}}\right)_{\text{reg}}$ into $\mathbb{Q}/\mathbb{Z}$ (resp. $H^2\left(\operatorname{Gal}\left(\overline{K}/L\right),C_{\overline{K}}\right)_{\text{reg}}$ into $\mathbb{Q}/\mathbb{Z}$). Thus for $a\in H^2\left(\operatorname{Gal}\left(\overline{K}/K\right),C_{\overline{K}}\right)_{\text{reg}}$, we have $\beta_2(a)=\operatorname{inv}_2(b)$, where $\varepsilon_2(b)=a$ (this is independent of the choice of b). Define

$$H^2\left(\operatorname{Gal}(L/K),C_L\right)_{\text{reg}}=\left\{a\in H^2\left(\operatorname{Gal}(L/K),C_L\right)\mid\operatorname{infl}a\in H^2\left(\operatorname{Gal}\left(\overline{K}/K\right),C_{\overline{K}}\right)_{\text{reg}}\right\}.$$

Then $n\beta_2(\operatorname{infl} a) = \beta_3(\operatorname{res}(\operatorname{infl} a)) = 0$ (note that the inflation-restriction sequence is exact), β_2 induces a homomorphism

$$\beta_1 \colon H^2\left(\mathrm{Gal}(L/K), C_L\right)_{\mathrm{reg}} \longrightarrow \frac{1}{n}\mathbb{Z}/\mathbb{Z}$$

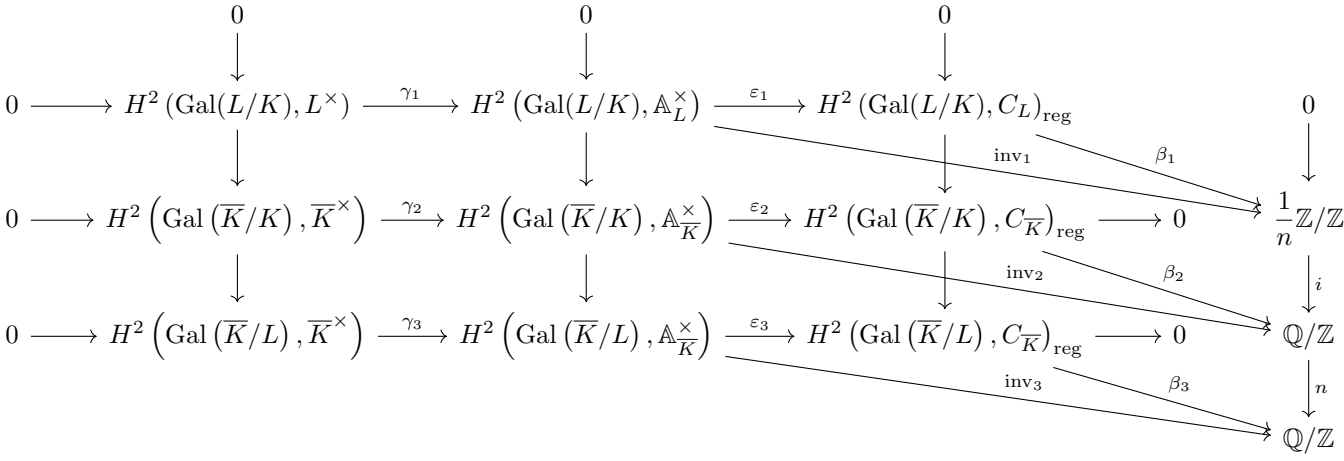
such that

$$\beta_1(a) = \beta_2(\operatorname{infl} a).$$

If $a = \varepsilon_1 b$ with $b \in H^2\left(\mathrm{Gal}(L/K), \mathbb{A}_L^\times\right)$ then

$$\beta_1(a) = \beta_2(\operatorname{infl} a) = \mathrm{inv}_2(\operatorname{infl} b) = \mathrm{inv}_1(b)$$

We conclude a new commutative (three-dimensional) diagram



in which i is the inclusion map, n is multiplication by n and the "bent" sequences are complexes, and the horizontal and vertical sequences are exact.

Let n_0 be the lowest common multiple of all the local degrees of L/K. Recall that

$$H^2\left(\mathrm{Gal}(L/K), \mathbb{A}_L^\times\right) \simeq \bigoplus_v \frac{1}{n_v}\mathbb{Z}/\mathbb{Z},$$

where $n_v = [L^v : K_v]$, we conclude $\mathrm{Im}(\mathrm{inv}_1) = \frac{1}{n_0}\mathbb{Z}/\mathbb{Z}$. Since $\mathrm{Im}(\mathrm{inv}_1) \subseteq \mathrm{Im}(\beta_1)$ and the second inequality gives $\left[H^2(\mathrm{Gal}(L/K), C_L)\right] \mid n$, we have

$$n \geq \left[H^2(\mathrm{Gal}(L/K), C_L)\right] \geq \left[H^2(\mathrm{Gal}(L/K), C_L)_{\mathrm{reg}}\right] \geq [\mathrm{Im}\,\beta_1] \geq [\mathrm{Im}(\mathrm{inv}_1)] \geq n_0.$$

Suppose first that L/K is cyclic. Recall the Frobenius elements $F_{L/K}(v)$, with orders the local degrees n_v, generate $\mathrm{Gal}(L/K)$. Hence the order of $\mathrm{Gal}(L/K)$ divides n_0. Since G is cyclic of order n, we must have $n = n_0$. Consequently

$$\beta_1 \colon H^2(\mathrm{Gal}(L/K), C_L)_{\mathrm{reg}} \xrightarrow{\sim} \frac{1}{n}\mathbb{Z}/\mathbb{Z}.$$

In particular, for cyclic cyclotomic L/K, the sequence

$$0 \longrightarrow H^2\left(\mathrm{Gal}(L/K), L^\times\right) \xrightarrow{\gamma_1} H^2\left(\mathrm{Gal}(L/K), \mathbb{A}_L^\times\right) \xrightarrow{\mathrm{inv}_1} \mathbb{Q}/\mathbb{Z}$$

is exact: if $\mathrm{inv}_1(b) = 0$, then $0 = \beta_1(\varepsilon_1 b)$, hence $\varepsilon_1 b = 0$, and therefore $b \in \mathrm{Im}\,\gamma_1$.

Recall that if $\alpha \in \mathrm{Br}(K)$, then α is split by a finite extension of K, i.e., $\alpha \in H^2\left(\mathrm{Gal}(L/K), L^\times\right)$ for some finite extension L/K. Combine with the inflation-restriction exact sequence, we conclude

$$H^2\left(\mathrm{Gal}\left(\overline{K}/K\right), \overline{K}^\times\right) = \bigcup_{L/K} \mathrm{infl}\, H^2\left(\mathrm{Gal}(L/K), L^\times\right).$$

Here L runs over all cyclic cyclotomic extensions of K. Similarly by the lemma in the Step 3 of the proof of reciprocity law, we get

$$H^2\left(\mathrm{Gal}\left(\overline{K}/K\right), \mathbb{A}_{\overline{K}}^\times\right) = \bigcup_{L/K} \mathrm{infl}\, H^2\left(\mathrm{Gal}(L/K), \mathbb{A}_L^\times\right).$$

Therefore the complex

$$0 \longrightarrow H^2\left(\mathrm{Gal}\left(\overline{K}/K\right), \overline{K}^\times\right) \xrightarrow{\gamma_2} H^2\left(\mathrm{Gal}\left(\overline{K}/K\right), \mathbb{A}_{\overline{K}}^\times\right) \xrightarrow{\mathrm{inv}_2} \mathbb{Q}/\mathbb{Z}$$

and

$$0 \longrightarrow H^2\left(\mathrm{Gal}\left(\overline{K}/L\right), \overline{K}^\times\right) \xrightarrow{\gamma_3} H^2\left(\mathrm{Gal}\left(\overline{K}/L\right), \mathbb{A}_{\overline{K}}^\times\right) \xrightarrow{\mathrm{inv}_3} \mathbb{Q}/\mathbb{Z}$$

are both exact. Therefore $\ker(\mathrm{inv}_2) = \ker(\varepsilon_2)$, so β_2 (and similarly β_3) are injective. They are also surjective, since local extensions of arbitrarily large degree exist, so inv_2 and inv_3 are surjective. Thus β_2, β_3 are isomorphisms.

Now return to an arbitrary finite Galois L/K. In the commutative exact diagram

$$\begin{array}{ccccccc}
0 & \longrightarrow & H^2(\mathrm{Gal}(L/K), C_L)_{\mathrm{reg}} & \xrightarrow{\mathrm{infl}} & H^2(\mathrm{Gal}\left(\overline{K}/K\right), C_{\overline{K}})_{\mathrm{reg}} & \xrightarrow{\mathrm{res}} & H^2(\mathrm{Gal}\left(\overline{K}/L\right), C_{\overline{K}})_{\mathrm{reg}} \\
 & & \downarrow{\scriptstyle \beta_1} & & \downarrow{\scriptstyle \beta_2} & & \downarrow{\scriptstyle \beta_3} \\
0 & \longrightarrow & \dfrac{1}{n}\mathbb{Z}/\mathbb{Z} & \longrightarrow & \mathbb{Q}/\mathbb{Z} & \xrightarrow{n} & \mathbb{Q}/\mathbb{Z}
\end{array}$$

the last two vertical maps are isomorphisms, hence so is β_1. Therefore

$$H^2(\mathrm{Gal}(L/K), C_L)_{\mathrm{reg}} \simeq \frac{1}{n}\mathbb{Z}/\mathbb{Z}$$

But this group is contained in $H^2(\mathrm{Gal}(L/K), C_L)$, whose order divides n. Hence

$$H^2(\mathrm{Gal}(L/K), C_L) \simeq \frac{1}{n}\mathbb{Z}/\mathbb{Z}$$

Letting $L \to \overline{K}$ we see that

$$H^2(\mathrm{Gal}\left(\overline{K}/K\right), C_{\overline{K}})_{\mathrm{reg}} = H^2(\mathrm{Gal}\left(\overline{K}/K\right), C_{\overline{K}}) \simeq \mathbb{Q}/\mathbb{Z}$$

Therefore, let L/K be arbitrary finite Galois extension, $H^2(\mathrm{Gal}(L/K), C_L)$ is cyclic of order n, and it has canonical generator $u_{L/K}$ with invariant $\frac{1}{n}$, i.e., $\beta_1(u_{L/K}) = \frac{1}{n}$. This element $u_{L/K}$ is called the **fundamental class** of the extension L/K.

Now we can prove the existence theorem. If H is an open subgroup of C_K, of finite index $[C_K : H]$, we say that H is **normic** if and only if there is an abelian extension L/K such that $H = \mathrm{N}_{L/K} C_L$.

First, two obvious remarks: If $H_1 \supset H$, and H is normic, then H_1 is normic (the field L corresponding to H_1 has a subfield L_1 corresponding to H). If H_1, H_2 are normic, so is $H_1 \cap H_2$, (take as the field the compositum $L_1 L_2$).

Lemma 3.1. Let n be a prime, and K a field not of characteristic n containing the nth roots of unity. Then every open subgroup H of index n in C_K is normic.

Proof. Suppose H is open in C_K with $[C_K : H] = n$. Let H' be the inverse image of H in $\mathbb{A}_K^\times$. Then H' is open in $\mathbb{A}_K^\times$, so there is a finite set $S \subset |K|$ such that $H' \supset \prod_{v \in S}(1) \times \prod_{v \notin S} \mathcal{O}_v^\times = U^S$. Furthermore, H is of index n in C_K, so $H' \supset {\mathbb{A}_K^\times}^n$. Therefore $H' \supset \prod_{v \in S} {K_v^\times}^n \times \prod_{v \notin S} \mathcal{O}_v^\times = E$, say. Thus $H = H'/K^\times \supset EK^\times/K^\times$. It follows that H is normic. □

Lemma 3.2. If L/K is cyclic and $H \subset C_K$, and if $\mathrm{N}_{L/K}^{-1}(H)$, is normic for L, then H is normic for K.

Proof. Write $H' \coloneqq \mathrm{N}_{L/K}^{-1}(H)$ and let M/L be the class field of H', i.e., $\mathrm{N}_{M/L} C_M = H'$. We claim that M is abelian over K, and $\mathrm{N}_{M/K} C_M \subset H$, so H is normic. That $\mathrm{N}_{M/K} C_M = \mathrm{N}_{L/K}\left(\mathrm{N}_{M/L} C_M\right) = \mathrm{N}_{L/K}(H') \subset H$ is clear. If one can prove M/K abelian, then by the previous "obvious remark" H is normic.

M is invariant under $\mathrm{Gal}(L/K)$ since H' is invariant under $\mathrm{Gal}(L/K)$, so M/K is normal, and so is Galois. The Galois group $E = \mathrm{Gal}(M/K)$ is a group extension,

$$1 \longrightarrow A = \mathrm{Gal}(M/L) \longrightarrow E = \mathrm{Gal}(M/K) \longrightarrow G = \mathrm{Gal}(L/K) \longrightarrow 1 \ ;$$

since $E/A \simeq G$ is cyclic, it is enough to show that $A = \mathrm{Gal}(M/L)$ is in the centre of E.

Let $\sigma \in E$ have image $\overline{\sigma} \in G$. We have the following picture

$$\begin{array}{ccc} M & \xrightarrow{\sigma} & M \\ | & & | \\ L & \xrightarrow{\overline{\sigma}} & \sigma L \end{array}$$

So for $x \in C_L$ and $y \in M$, $(\psi(\overline{\sigma}x))(\sigma y) = \sigma(\psi(x)(y))$ where $\psi\colon C_L \to A$ is the Artin map. Equivalently $\psi(\overline{\sigma}x) = \sigma\psi(x)\sigma^{-1}$.

To show that A is in the center, it is enough to check that $\psi(x) = \sigma\psi(x)\sigma^{-1} = \psi(\sigma x)$ for all $x \in C_L$ and $\sigma \in E$. Now $\psi\colon C_L \to A$ has kernel H', so we want to check that $\sigma x/x \in H'$, which is clear since $\mathrm{N}(\sigma x/x) = 1$. □

For the proof of existence theorem, we can only treat the case in which the index is prime to the characteristic. We use induction on the index of H. If the index is 1 everything is clear. Now let n be a prime dividing the index. Adjoin the nth roots of unity to K to get K', and replace H by $H' = \mathrm{N}_{K'/K}^{-1}(H)$. We reduce to consider H' by Lemma 3.2, and may assume $[C_{K'} : H'] = [C_K : H]$, otherwise H' is normic by induction hypothesis. So n divides $[C_{K'} : H']$.

Take H_1' so that $H_1' \supset H'$ and $[C_{K'} : H_1'] = n$, then H_1' is normic by Lemma 3.1. Let L be its class field, i.e., $H_1' = \mathrm{N}_{L/K'} C_L$. Put $H'' = \mathrm{N}_{L/K'}^{-1}(H')$. Then

$$[C_L : H''] < [C_{K'} : H'] = [C_K : H].$$

(For $C_L/H'' \xrightarrow{\mathrm{N}_{L/K'}} C_{K'}/H'$ is an injection, whose image is H_1'/H', properly contained in $C_{K'}/H'$, since $n > 1$ a prime.)

Hence H'' is normic by induction hypothesis; L/K' is cyclic, so H' is normic by Lemma 3.2.

4 Examples

To conclude this note, we provide some examples.

Example 4.1 (Hilbert class field)**.** Let K be a number field. The kernel of the canonical surjection $\mathbb{A}_K^\times/K^\times \to \mathrm{Cl}(K)$ is an open subgroup of finite index of $\mathbb{A}_K^\times/K^\times$. The finite abelian extension $\widetilde{K}$ of K that corresponds to this subgroup through class field theory is called the **Hilbert class field** of K.

By class field theory we have

$$\mathrm{Cl}(K) \cong \mathrm{Gal}\left(\widetilde{K}/K\right).$$

Every prime ideal $\mathfrak{p}$ of K is unramified in $\widetilde{K}$, and the class of $\mathfrak{p}$ in $\mathrm{Cl}(K)$ is mapped to $\mathrm{Frob}_\mathfrak{p} \in \mathrm{Gal}\left(\widetilde{K}/K\right)$. Therefore, we have

$$\mathfrak{p} \text{ is a principal ideal} \iff \mathfrak{p} \text{ is totally decomposed in } \widetilde{K}.$$

Example 4.2. Let L/K be a quadratic extension of local fields. Local class field theory gives an isomorphism

$$K^\times / \operatorname{N}_{L/K}\left(L^\times\right) \cong \operatorname{Gal}(L/K) \cong \{\pm 1\}.$$

This induces a quadratic character

$$\eta\colon K^\times \longrightarrow \{\pm 1\}, \quad \operatorname{Ker}(\eta) = \operatorname{N}_{L/K}\left(L^\times\right).$$

When $K = \mathbb{R}$ and $L = \mathbb{C}$, this is precisely $\operatorname{sgn}\colon \mathbb{R}^\times \to \{\pm 1\}$.

Let L/K be a quadratic extension of number fields. Global class field theory gives an isomorphism

$$C_K / \left(\operatorname{N}_{L/K} C_L\right) \cong \operatorname{Gal}(L/K) \cong \{\pm 1\}$$

This induces a quadratic character

$$\eta\colon \mathbb{A}_K^\times / K^\times \longrightarrow \{\pm 1\}, \quad \operatorname{Ker}(\eta) = \operatorname{N}_{L/K} C_L,$$

such that for every non-split place $v \in |K|$, η_v is the local quadratic character above.

Such quadratic characters occur frequently in automorphic forms.

Example 4.3. Recall that a locally compact abelian group is essentially the same as knowing its character group. By class field theory,

$$\begin{aligned}
\{\text{characters of } \operatorname{Gal}(L/K)\} &= \bigcup_L \{\text{characters of } \operatorname{Gal}(L/K)\} \\
&= \bigcup_H \{\text{characters of } C_K/H\} \\
&= \{\text{characters of finite index of } C_K\}.
\end{aligned}$$

Let L/K be an abelian extension of number fields. Let H be the open subgroup of finite index of C_K corresponding to L. Then, we have

$$\zeta_L(s) = \prod_\chi L(s, \chi),$$

where χ runs through all the characters of finite abelian group C_K/H. The Dedekind ζ-function $\zeta_L(s)$ can be computed by easy-to-understand Hecke L-function $L(s, \chi)$.

Example 4.4. Recall that a Hecke character χ is **unramified** at v if χ_v is trivial on $\mathcal{O}_v^\times$. In this case, $\chi_v(\pi_v)$ does not depend on the choice of uniformizing element $\pi_v \in K_v$. One natural question is what is the relation between it and "ramification index" in the algebraic number theory.

Under local reciprocity law, $\mathcal{O}_v^\times$ corresponds to the inertia group and $1+\mathfrak{p}_v^n$ corresponds to the ramification groups in the upper numbering. The $\mathrm{Gal}\left(\overline{K}/K\right)$-set is called **unramified** if the I_K-action is trivial.

Let $\chi_v\colon K_v^\times \to \mathbb{C}^\times$ be a continuous character of finite order. Then there exists m such that $\chi_v(x)^m = 1$ for all $x \in K_v^\times$, i.e., $\chi_v\left(K_v^\times\right) \subset \mu_m = \{z \in \mathbb{C}^\times \mid z^m = 1\}$. It follows that $K_v^\times / \operatorname{Ker}\chi_v \xrightarrow{\sim} \chi_v\left(K_v^\times\right)$, $\operatorname{Ker}\chi_v$ has finite index. Through local class field theory there exists L such that

$$\mathrm{N}_{L/K_v}\left(L^\times\right) = \operatorname{Ker}\chi_v,$$

and we have

$$\mathrm{Gal}(L/K_v) \cong K_v^\times / \operatorname{Ker}\chi_v \cong \chi_v\left(K_v^\times\right).$$

Define the character $\overline{\rho}$ on $\mathrm{Gal}\left(L/K_v\right)$ through

$$\overline{\rho}\left(\theta_{L/K_v}(x)\right) = \chi_v(x)$$

where θ_{L/K_v} is the local reciprocity map. It is well-defined and injective, because $\chi_v(x) = 1$ implies that $x \in \operatorname{Ker}\chi_v$ implies that $\theta_{L/K_v}(x) = 1$. Define the character ρ on $\mathrm{Gal}\left(\overline{K_v}/K_v\right)$ through

$$\mathrm{Gal}\left(\overline{K_v}/K_v\right) \xrightarrow{\text{restriction}} \mathrm{Gal}\left(L/K_v\right) \xrightarrow{\overline{\rho}} \mathbb{C}^\times.$$

Therefore,

$$\#\rho I\left(\overline{K}_v/K_v\right) = \#\overline{\rho} I\left(L/K_v\right) = \#I\left(L/K_v\right) = e(L/K_v).$$

On the other hand, note that $\theta_{L/K_v}\left(\mathcal{O}_v^\times\right) = I\left(L/K_v\right)$, we conclude

$$\chi_v\left(\mathcal{O}_v^\times\right) = \overline{\rho}\left(\theta_{L/K_v}\left(\mathcal{O}_v^\times\right)\right) = \overline{\rho} I\left(L/K_v\right).$$

Consequently,

$$e\left(L/K_v\right) = \#\rho I\left(\overline{K_v}/K_v\right) = \#\chi_v\left(\mathcal{O}_v^\times\right).$$

χ_v is unramified if and only if L/K_v is unramified.